Fourier Series

It is a method to break a complex periodic function into infinite sum of sines and cosines which are harmonically related

Periodic Signal $f(t) = f(t+T)$

where T is time period $\frac{1}{T}$ = frequency

$$\left(\omega = \frac{2\pi}{T}\right)$$

$$f(t) = \underbrace{a_0}_{\text{DC component}} + \sum_{n=1}^{\infty} a_n \cos n\omega t + \sum_{n=1}^{\infty} b_n \sin n\omega t$$

Some Integration

$$\int_0^T \sin \omega t = 0 \qquad \int_0^T \cos \omega t = 0$$

$$\int_0^T \sin^2 \omega t \, dt = \frac{T}{2} = \frac{2\pi}{2} = \pi$$

$$\int_0^T \cos^2\omega t = \frac{T}{2} \qquad \int_0^T \sin n\omega t \cos n\omega t = 0$$

$$\int_0^T \sin n\omega t \cos m\omega t = \begin{cases} 0 & n \neq m \\ T/2 & n = m \end{cases}$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t$$

$$+ \sum_{n=1}^{\infty} b_m \sin n\omega t$$

$$\int_0^T f(t) \cos m\omega t \, dt = \overset{0}{\cancel{\int_0^T a_0 \cos m\omega t \, dt}}$$

$$+ \int_0^T \sum_{n=1}^{\infty} a_n \cos n\omega t \cos m\omega t \, dt$$

$$+ b_n \overset{0}{\cancel{\int_0^T \sum_{n=1}^{\infty} \cos m\omega t \sin n\omega t \, dt}}$$

$$\int_0^T f(t) \cos n\omega t = a_n \int_0^T \cos^2 n\omega t$$

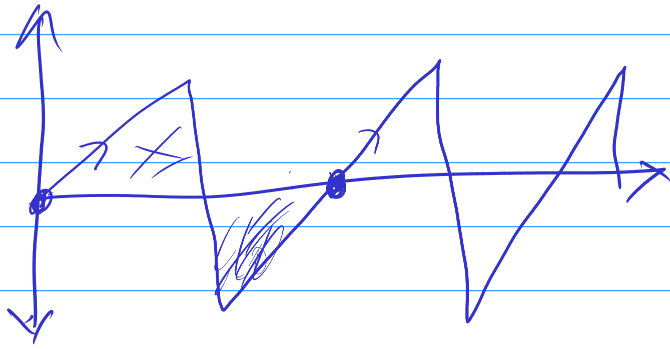
$$\int_0^T f(t) \cos n\omega t = \frac{T a_n}{2}$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos n\omega t$$

$$b_n = \frac{2}{T}\int_0^T f(t)\sin n\omega t$$

$$a_0 = \int_0^T f(t)\,dt$$

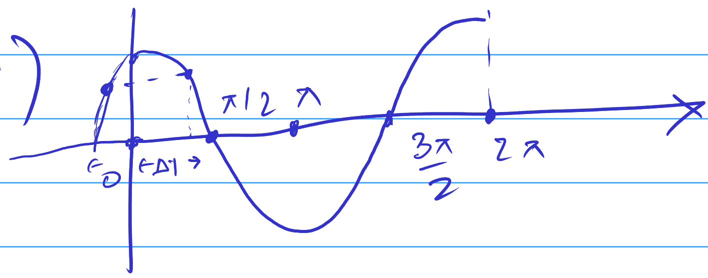
Symmetric about x axis



$a_0 = 0$

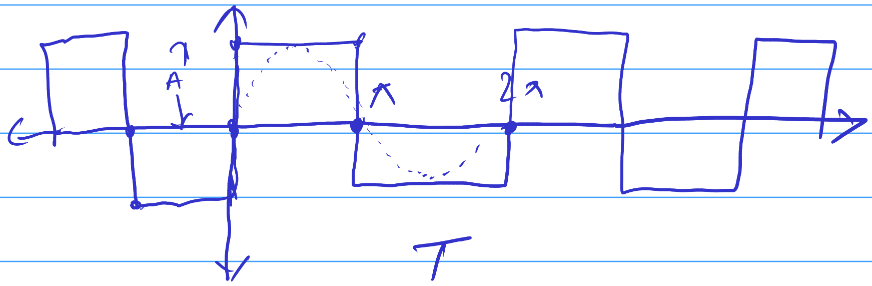
$\cos t \leftarrow$ it is an even function

$f(T) = f(-T)$



For an even function no sin terms will be there $b_n = 0$

For odd functions no cosine terms will be there $a_n = 0$



$\Leftarrow$ odd function

$a_n = 0$

$a_0 = 0$

$$b_n = \frac{2}{T} \int_0^T f(t) \sin n\omega t \, d\omega t$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(t) \sin n\omega t \, d\omega t$$

$$b_n = \frac{1}{\pi} \left[\int_0^{\pi} f(t) \sin n\omega t \, d\omega t + \int_{\pi}^{2\pi} f(t) \sin n\omega t \, d\omega t \right]$$

$$b_n = \frac{1}{\pi} \times 2 \left[\int_0^{\pi} A \sin n\omega t \, d\omega t \right]$$

$$b_n = \frac{2A}{\pi} \left[-\frac{\cos n\omega t}{n} \right]_0^{\pi}$$

$$= +\frac{2A}{\pi n}\left[+1+1\right]$$

$$= \frac{4A}{\pi n}$$

$$\text{Square wave} = \sum_{n=1}^{\infty} \frac{4A}{\pi n} \sin n\omega t$$

xxx

odd function

$a_0 = 0$

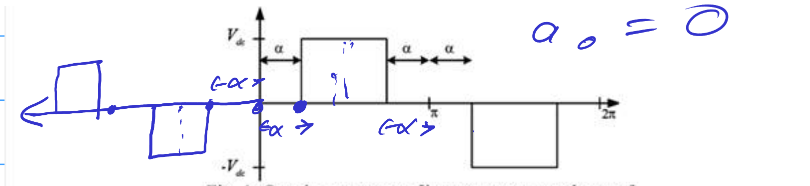
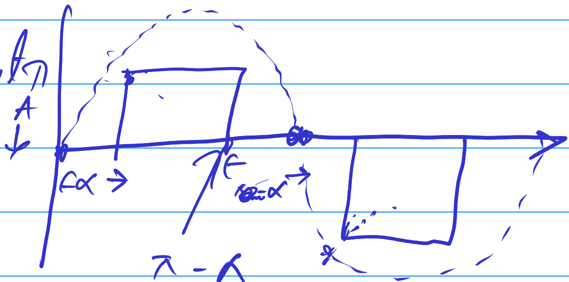


Fig. 1. Quasi square wave [image source: gradeup.co]

$a_n = 0, \; a_0 = 0$ $\quad b_n$

$$b_n = \frac{2}{2\pi}\int_0^{2\pi} f(t)\sin n\omega t \, d\omega t$$

$$b_n = \frac{1}{\pi}\times 2\times\int_{\alpha}^{\pi-\alpha} A \sin n\omega t \, d\omega t$$



$$b_n = \frac{2A}{\pi}\left[\frac{-\cos n\omega t}{n}\right]_{\alpha}^{\pi-\alpha}$$

$$= \frac{-2A}{\pi n}\left[\cos n(\pi-\alpha) - \cos n\alpha\right]$$

$n =$ even $\quad \cos n(\pi-\alpha) = \cos n\alpha$

$n =$ odd $\quad \cos n(\pi-\alpha) = -\cos n\alpha$

$$= \frac{2A}{\pi n}\times 2\cos n\alpha = \frac{4A\cos n\alpha}{\pi n}$$

Quasi Square Wave can be written as

$$f(t) = \sum_{n=1}^{\infty} \frac{4A \cos n\alpha}{\pi n} \sin n\omega t$$